

PUBH 7430 – Assignment 2

On this assignment, as in all assignments, inclusion of raw R/SAS output (e.g., screenshots of output or R Markdown raw output) is not acceptable. Instead, you should extract the relevant information and summarize it in your response. For example, instead of including the raw test output like in the top half of slide 25 of Lecture 3, you should summarize the findings in a sentence like in the bottom half of slide 25 of Lecture 3. Any tables and plots prepared from statistical output should be appropriate for inclusion in a scientific report, with all statistics rounded to a reasonable number of significant digits and labeled as needed. Obtaining the correct results from an analysis is only a portion of each exercise. It is also very important to write clearly about what your results mean. This includes presenting your results in a way that clearly answers the question and places little burden on the reader.

*You should submit your assignment on Canvas as a PDF file. While no code should be included in the body of the assignment, you **must include your code as an appendix** at the end of the assignment. Keep in mind that while working together on homework assignments is permitted, each student is expected to independently write up homework assignments, including any code. You are responsible for reading the generative AI (genAI) policy in the syllabus. When genAI tools are used for tasks beyond grammar improvement and debugging code, you should include a disclosure statement in the assignment. For example: "For this assignment, I used [list of genAI tools used in assignment] to [list the task(s)] for [list of the portions of the assignment]."*

Background

This dataset is on dental caries (tooth decay) caused by mutans streptococci (MS).

This study recruited 151 persons with gum disease (elevated oral MS levels) at a Midwestern research university and randomly assigned each study participant to one of three treatment groups. The treatment under study was a gum with an active drug (xylitol). Two control groups were included: one received gum with sorbitol but no xylitol, and one received no gum. Participants in the gum groups were asked to chew the gum three times daily for a minimum of five minutes each time and carry out their usual oral hygiene (tooth brushing, mouthwash, etc.). Participants in the no-gum group were asked to perform their usual oral hygiene only.

Subjects rinsed their mouths twice daily for 14 days with a 0.12% CHZ gluconate mouthrinse. They were then asked to follow their assigned treatment for three months. Participants had their oral levels of mutans streptococci measured four times: at baseline after the 14 days of CHZ gluconate mouthrinse (after which they were randomly allocated to one of the three treatment groups), one week later, one month later, and three months later. The four *mut** variables are the levels of oral mutans streptococci at the four time periods in units of log(cfu/ml) (where cfu = "colony forming units", a count of blotches on a standard sized petri dish after standard preparation). Missing MS levels are recorded as 'NA' in the R data file ("gumR2026.csv" on Canvas) and '.' in the SAS data file ("gumSAS2026.csv" on Canvas).

The variables *dmft* and *dmfs* reflect the general condition of the teeth: *dmft* stands for diseased/missing/filled teeth and *dmfs* stands for diseased/missing/filled surfaces. (There are five surfaces per tooth: the four sides and the chewing surface.)

Questions

For questions 1 and 2, we are interested in comparing the treatment effect for active treatment vs. gum placebo. For these questions, you can exclude the patients who received no gum. For question 3, you should include all patients in your analysis.

1. **Treatment effects, part 1 [14 pts].** Suppose we are interested in testing the efficacy of the **active treatment versus gum placebo** at three months post-baseline (outcome *mut3mos*).
 - (a) [2 pts] Fit a Gaussian GLM with identity link (i.e., a linear regression) to assess the effect of treatment on mutans streptococci levels at three months, and write a sentence summarizing the estimated treatment effect.
 - (b) [3 pts] Fit the same model, but include mutans streptococci level at baseline as a predictor. Write a sentence summarizing the estimated treatment effect from this model. How much, if at all, do your conclusions change? What is the purpose of including baseline MS level in this model (e.g., confounder, precision variable, effect modifier)?
 - (c) Fit a model to assess whether the effect of treatment differs by baseline MS level.
 - i. [2 pts] Write down the mean model in terms of β . Summarize in words what the coefficient of the treatment term (i.e., the “main effect” of treatment) in this model is estimating. Is this the same or different than what’s being estimated by the treatment coefficients in parts (a) and (b)? Explain.
 - ii. [1 pt] What role does baseline MS level play in this model (e.g., potential confounder, potential precision variable, potential effect modifier)?
 - iii. [1 pt] Reviewing your model results, is there evidence of the treatment effect differing by baseline MS level?
 - iv. [2 pts] Using this model, estimate the treatment effect (active treatment vs. gum placebo) for a participant with a baseline MS level of 5.43 log(cfu/ml), the mean baseline MS level in the sample. Write the requested treatment effect in terms of the β ’s using your specified mean model, and then calculate and report the estimate and 95% confidence interval.
 - v. [2 pts] Use a joint hypothesis test to test whether there is any effect of treatment on 3-month MS levels in this model. Clearly state the null hypothesis in terms of the β ’s in your specified mean model and report the p-value from performing the test.
 - vi. [1 pt] Why is the confidence interval for the “main effect” of treatment relatively wide? How would you expect the confidence interval width of the estimated treatment effect to change for baseline MS levels closer to vs. farther from the center of the observed data?
2. **Treatment effects, part 2 [8 pts].** Suppose that your colleagues in the Dental School tell you that having a mutans streptococci level greater than 5.5 log(cfu/ml) is known to be associated with particularly poor dental outcomes. (Note: When fitting logistic regression models using PROC GENMOD in SAS, SAS models the odds of a value of 0 (rather than the usual value of 1). Thus, you almost always want to tell SAS to do the opposite by specifying the DESCENDING option (e.g., proc genmod data=gum descending;).)
 - (a) [3 pts] Define a binary outcome variable indicating whether or not a patient’s MS level exceeds 5.5 log(cfu/ml) at one month post-baseline. Fit a model to compute the odds ratio of this outcome associated with treatment (xylitol vs. gum placebo). Write down the mean model in terms of β . Summarize your conclusions in a sentence or two suitable for inclusion in a scientific publication.

- (b) [3 pts] Using the binary outcome variable defined in 2(a), now fit a model to compute the relative risk of this outcome associated with treatment (xylitol vs. gum placebo). Write down the mean model in terms of β . Summarize your conclusions in a sentence or two suitable for inclusion in a scientific publication.
- (c) [2 pts] Would you expect the results from 2(a) and 2(b) to be similar? If so, explain why you expect similar results. If not, when would you expect similar results?
3. **Regression and log transformation [18 pts]**. Suppose we are interested in exploring the association between the condition of the teeth (as reflected by the variable *dmfs*) and age.
- (a) [2 pts] Produce histograms showing the distributions of *dmfs* and $\log(dmfs)$. Summarize what you see in your plots.
- (b) Fit the following four statistical models with *dmfs* as the outcome variable and age as the predictor. For each model, write down the mean model in terms of β , and summarize your conclusions about the association between *dmfs* and age in a sentence or two suitable for inclusion in a scientific publication. When appropriate, provide an interpretation of the exponentiated coefficient for age, instead of the non-exponentiated coefficient.
- [3 pts] A Gaussian GLM with identity link (i.e., a linear regression).
 - [3 pts] A Gaussian GLM with identity link, using log-transformed values of *dmfs*. (Given the zero values, you will need to make a small modification to the data to fit this model. For example, you can replace the zero values for *dmfs* with 0.1.)
 - [3 pts] A Gaussian GLM with log link (use non-transformed *dmfs* values). (Note: In R, there will be an error about finding initial values if you try to fit this model due to the person with a zero value of *dmfs*. A workaround is to fit the same model but with the data slightly perturbed (i.e., replace the zero value with 0.1). SAS doesn't have this issue.)
 - [3 pts] A Poisson GLM with log link (again, use non-transformed values).
- (c) [2 pts] For the four models fit in 3(b), which two models have the most similar results? Explain why these two models would be expected to produce similar results. Which model's results are the most different? Explain why this model would be expected to produce very different results.
- (d) [2 pts] Produce plots of standardized Pearson residuals versus fitted values for each model you fit in 3(b). Which of the four plots is most concerning in terms of model fit? Explain.